scientific reports

OPEN

Reliable equalization aided long-distance OFDM-FSO performance analysis over Gamma–Gamma turbulence with thermal and background noise mitigation

R. Balakrishnan[1]✉ & S. Senthilkumar[2]

Free Space Optical Orthogonal Frequency Division Multiplexing (FSO- OFDM) has emerged as a powerful distinct transmission framework for next-generation wireless links; here the performance of FSO system using OFDM in Low Earth Orbit/Medium Earth Orbit (LEO/MEO) inter-satellite communication is investigated under the combined effects of thermal, background noise and solar storm disturbances over a Gamma–Gamma (GG) channel model. An impact of channel impairments are analysed for BPSK, QPSK and M-ary (M = 64,128,256) QAM transmission. A bit error rate in the received signal is analysed for different refractive index with weak and strong turbulence conditions. To increase the transmission link distance and reduce the bit error rate, we employed zero forcing equalization techniques at the receiver. We examine the different transmission channel length from 1000 to 11000 m. Lower order PSK modulation schemes such as BPSK and QPSK indorse reliable performance for link ranges up to 7000 m due to their robustness to turbulence and noise. The simulation results substantiate that in weak turbulence OFDM-FSO system, 256—QAM achieves high spectral efficiency with a BERvalue of 10^{-12} at 23 dB SNR. Under severe turbulence, BPSK is optimal up to 7000 m with a BERof 10^{-12} at 22 dB SNRin weak, and 27 dB in strong turbulence, whereas for long distance up to 11000 m, 256—QAM becomes more efficient, reaching a BERof 10^{-6} at 27.5 dB SNR. The analysis further demonstrate that for an identical symbol rate, the BER achieved at 11000 m in weak turbulence closely bouts the BER obtained at 9000 m when the channel experiences strong turbulence like solar storm. Furthermore, the developed equalization framework exhibits superior performance compared to traditional attitudes, demonstrating its capability to sustain stable long-range OFDM-FSO communication for LEO/MEO inter-satellite links.

Keywords OFDM, FSO, Gamma–Gamma turbulence, BPSK (binary phase shift keying), QPSK (quadrature phase shift keying), QAM (quadrature amplitude modulation), Refractive index, Equalization

In recent years, the rapid growth of high speed video conferencing, ultra-fast internet with amended quality of service has driven an exponential demand for channel capacity and bandwidth. In free space optical communication (FSOC), optical signals are transmitted over the atmospheric free space as the propagation medium. Compared with RF transmission, FSO provides several advantages, by combining the spectral efficiency and multipath springiness of OFDM with ultra—wide bandwidth prospective of optical carriers, FSO-OFDM enables high capacity, multi- giga bit data rates, low latency communication over licence-free optical channels and relatively low installation cost[1]. As a result, FSO is increasingly viewed as balancing tools to RF and millimetre-wave structures for high-speed data, voice, and video communication. Its architecture is particularly gorgeous for long-range and high-data rate applications, where conventional RF systems face spectrum congestion and hardware limitations. Nevertheless, FSO communication links are significantly exaggerated by atmospheric turbulence and adverse weather conditions including rain, snow, haze, fog, lighting and thunderstorms, the fluctuations in irradiance due to the sun (solar storm)[2] and sky, thermal expansion pointing errors. The FSOC

[1]Department of Electronics and Communication Engineering, Kings College of Engineering, Punalkulam, pudhukottai, Tamil Nadu 613303, India. [2]Department of Electronics and Communication Engineering, E.G.S. Pillay Engineering College, Nagapattinam, Tamil Nadu 611002, India. ✉email: rbala.krish2011@gmail.com